A

Compound-class distribution across cohorts

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	Other
MatterGen Base	48.8	14.8	0.9	1.7	2.1	4.2	2.1	3.8	2.5	4.6	14.5
MatterGen FT	55.0	0.0	1.6	0.1	17.4	0.0	2.5	5.4	12.8	0.0	5.1
Con-CDVAE Base	14.4	21.0	1.9	14.4	1.6	14.0	0.3	4.3	0.7	13.5	13.8
Con-CDVAE FT	18.6	0.0	4.8	1.8	14.2	0.0	27.7	9.5	6.3	0.0	16.9
CrystalFormer Base	28.9	11.7	24.5	8.8	2.4	4.2	0.7	3.3	2.0	0.4	13.1
CrystaLLM Base	29.6	45.9	0.0	0.0	0.0	12.2	8.2	0.0	4.1	0.0	0.0
Crys-JEPA Base	30.3	16.3	1.7	13.2	4.7	11.3	2.2	4.0	5.0	3.1	8.3
Crys-JEPA Ref.	76.1	0.0	20.2	0.3	0.0	0.1	0.4	0.9	0.1	0.0	1.9

Share of cohort (%)

B

Fine-tuning and refinement shifts in compound classes

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	Other
MatterGen Fine-tuned − Base	+6.2	-14.8	+0.7	-1.5	+15.3	-4.2	+0.4	+1.6	+10.3	-4.6	-9.5
Con-CDVAE Fine-tuned − Base	+4.2	-21.0	+2.9	-12.7	+12.6	-14.0	+27.4	+5.3	+5.6	-13.5	+3.1
Crys-JEPA Refined − Base	+45.8	-16.3	+18.5	-12.8	-4.7	-11.2	-1.8	-3.1	-4.9	-3.1	-6.4

Change from baseline (percentage points)

C

Unique reduced formulas

	Unique reduced formulas (%)
MatterGen Base	84.0% (1392/1657)
MatterGen FT	29.9% (412/1376)
Con-CDVAE Base	98.8% (1842/1864)
Con-CDVAE FT	44.2% (574/1299)
CrystalFormer Base	66.5% (720/1083)
CrystaLLM Base	34.7% (34/98)
Crys-JEPA Base	98.1% (1730/1764)
Crys-JEPA Ref.	56.2% (867/1543)

D

Unique formula–space-group pairs

	Unique formula–space-group pairs (%)
MatterGen Base	96.5% (1599/1657)
MatterGen FT	70.2% (966/1376)
Con-CDVAE Base	99.2% (1849/1864)
Con-CDVAE FT	44.2% (574/1299)
CrystalFormer Base	86.7% (939/1083)
CrystaLLM Base	52.0% (51/98)
Crys-JEPA Base	98.2% (1733/1764)
Crys-JEPA Ref.	58.4% (901/1543)